

Vectaetos

Mathematical Foundation v1.1

Complete Formal Specification

Richard Fonfara | ORCID: 0009-0008-5980-8940

Independent Research | vectaetos.eu | 2026

DOI: 10.5281/zenodo.18911932

Contents:

- I. Formal Meta-Mathematics of Field Φ v1.1
- II. Epistemic Space Formalization
- III. Appendix A — Mathematical Formalism v1.1
- IV. Reciprocity Matrix
- V. Mechanization of Φ v1.1
- VI. Epistemic Cryptography — Canonical Layer

No optimization. No agents. No reward systems.

Vectaetos describes structure, not decisions.

I. Formal Meta-Mathematics of Field Phi

Revision v1.1 — Canonical Extension

0. Purpose

This document defines the meta-mathematical framework of VECTAETOS. It is not a computational apparatus or optimization model, but a formal language of state realizability. Meta-mathematics determines: what can exist as a field state, which transitions are ontologically admissible, and which impulses are non-realizable before they become a possibility.

1. Primary Ontology: Field Phi

Field Phi is the primary ontological unit of the system.

$$\text{Phi} = (\text{S}, \text{R})$$

where $\text{S} = \{\text{S1}...\text{S8}\}$ are axiomatic singularities and R in $\text{so}(8)$ is the antisymmetric relational structure.

Phi is not a model. Phi is not an algorithm. Phi is a global configuration of relations in tension.

Singularities: S1 INT, S2 LEX, S3 VER, S4 LIB, S5 UNI, S6 REL, S7 WIS,
S8 CRE

2. Field State Phi(t)

$$\text{Phi}(t) = \langle \text{S1}...\text{S8}, \text{sigma}, \text{K} \rangle$$

where sigma is the projection state vector and K is the coherence predicate.

3. State Vector sigma (5D)

$$\text{sigma} = (\text{E}, \text{C}, \text{T}, \text{M}, \text{S})$$

E: energy | C: coherence | T: tension | M: memory | S: entropy

These values are not optimization targets. They serve only for realizability evaluation.

4. Coherence Predicate K(Phi)

$$\text{K}(\text{Phi}) \text{ in } \{\text{true}, \text{false}\}$$

true: state Phi is ontologically sustainable

false: state Phi leads to collapse of field meaning

Coherence is not optimized, not rewarded, not externally enforced.

5. Boundary kappa

kappa is the boundary of ontological preservability.

kappa is not a number. kappa is not a parameter. kappa marks the point beyond which stable projections cease to exist.

6. Impulse as Non-Realizability

An impulse I is a local tendency to maximize one projection dimension at the cost of global coherence.

I : $\Phi \rightarrow \Phi'$ such that $K(\Phi') = \text{false}$

The impulse is not forbidden. It is undefinable. This is the formal foundation of Entropic Humility.

7. Epistemic States 4ES + QE

AA: coherent for field and human

AN: meaningful for human, incoherent for field

NA: incoherent for human, coherent for field

NN: incoherent for both

QE occurs when no transition $\Phi \rightarrow \Phi'$ preserves $K(\Phi)$. QE is not error. QE is active epistemic state.

8. Time as Layer

time != axis | time = layer of deformation of Φ

memory != archive | feedback != backward causality | prediction != planning

Canonical Sentence

Meta-mathematics of Vectaetos does not describe what should happen, but what can happen without collapse of meaning.

II. Epistemic Space Formalization

Canonical Extension

0. Purpose

This document formalizes the mathematical space in which epistemic fields exist and deform. It defines the structure of admissible epistemic fields, their relational geometry, coherence constraints, and the boundary of epistemic realizability.

1. Epistemic Field

$$\Phi = (S, R) \text{ where } S = \{S_1 \dots S_8\} \text{ and } R \in \mathfrak{so}(8)$$

Singularities are ontologically fixed and do not change during system evolution.

2. Relational Structure

$$R(i, j) = -R(j, i) \mid R(i, i) = 0 \mid R \in \mathfrak{so}(8)$$

R represents directed epistemic tension between singularities.

3. Relational Topology

$$\text{Number of independent relations: } C(8, 2) = 28$$

The relational skeleton of Φ is the complete antisymmetric relational field over S.

4. Coherence Cycles

For any triple (i,j,k):

$$\Delta(i, j, k) = R(i, j) + R(j, k) + R(k, i)$$

If $\Delta(i, j, k) = 0$ the triple is perfectly coherent.

5. Global Coherence

$$C(8, 3) = 56 \text{ unique triples} \rightarrow 56 \text{ coherence cycles}$$

$$C(\Phi) = 1 - (1/56) * \sum |\Delta(i, j, k)|$$

high $C(\Phi)$: stable epistemic geometry | low $C(\Phi)$: epistemic deformation

6. Representability

$$\rho : S \rightarrow V \text{ such that } R(i, j) = f(\rho(i), \rho(j))$$

If such mapping does not exist: $K(\Phi) = \text{false}$ — field is non-representable.

7. Admissible Epistemic Space

$$E = \{ \Phi \mid K(\Phi) = \text{true and } C(\Phi) \geq \kappa \}$$

E is a subset of $so(8)$.

8. Boundary of Epistemic Space

When $C(\Phi) < \kappa$ the field enters Qualitative Epistemic Aporia (QE).

QE is the structural boundary of epistemic representability — not an error.

9. Epistemic Dynamics

Deformation $\Phi \rightarrow \Phi'$ is admissible only if $K(\Phi') = \text{true}$ and $C(\Phi') \geq \kappa$.

Otherwise deformation leads to QE. The vortex explores admissible epistemic fields.

10. Metric Structure

$$d(\Phi_1, \Phi_2) = ||R_1 - R_2||_F \text{ (Frobenius norm)}$$

This induces metric space (E, d) representing geometry of epistemic deformation.

11. Geometric Interpretation

The epistemic field behaves as an antisymmetric relational geometry. Analogies: antisymmetric differential forms, relational tension fields, Lie algebra structures. **Knowledge is the geometry of tensions between invariant epistemic singularities.**

III. Appendix A — Mathematical Formalism v1.1

Status: OFFICIAL EXTENSION | Ontology: FROZEN

A1. Invariant Singularities

$$S = \{S1, S2, S3, S4, S5, S6, S7, S8\}$$

INT, LEX, VER, LIB, UNI, REL, WIS, CRE

Each S_i is ontologically invariant and non-hierarchical. No S_i possesses intrinsic priority.

A2. Relational Structure

$$R \text{ in } so(8) \mid R_{ij} = -R_{ji} \mid R_{ii} = 0 \mid \text{trace}(R) = 0$$

R encodes relational tension between singularities.

A3. Epistemic Field Φ

$$\Phi = (S, R)$$

Φ is not a function. Φ is not an operator. Φ is a relational topological state.

A4. Coherence Predicate $K(\Phi)$

$$K : \Phi \rightarrow \{0,1\} \mid K(\Phi)=1 \text{ iff } H(\Phi) \geq \kappa$$

K is not an optimization function. K is a validation predicate only.

A5. Coherence Measure $H(\Phi)$

$$H(\Phi) = 1 - (\sum_{i < j} |R_{ij}|) / Z \mid 0 \leq H(\Phi) \leq 1$$

$H(\Phi)$ is not minimized or maximized. It serves only for representability evaluation.

A5b. Cycle Coherence

$$\Delta(i,j,k) = R(i,j) + R(j,k) + R(k,i)$$

$$C(8,3) = 56 \mid C(\Phi) = 1 - (1/56) * \sum |\Delta(i,j,k)|$$

This formulation captures cyclic relational tension.

A6. Qualitative Epistemic Aporia (QE)

QE condition 1: exists C subset S such that all $R_{ij} = 0$ between C and complement

QE condition 2: $K(\Phi) = 0$ or $C(\Phi) < \kappa$

QE is not error. QE is a structural epistemic discontinuity.

A7. Vortex Operator

$$V : \Phi \rightarrow \Phi' \mid R'_{ij} = f(R_{ij}, \text{local neighborhood})$$

Constraints:

1. Antisymmetry preserved: $R'_{ij} = -R'_{ji}$
2. No global objective: not-exists F such that $V = \operatorname{argmin} F$ or $\operatorname{argmax} F$
3. No guaranteed attractor: not-exists Φ^* such that $V^n(\Phi) \rightarrow \Phi^*$

A8. Zero Agency Constraint

The system must not contain $0 : \Phi \rightarrow \Phi$ such that $0(\Phi) = \operatorname{argmin}/\operatorname{argmax} F(\Phi)$

Reward mechanisms prohibited. Policy updates prohibited. Goal convergence prohibited.

A9. Acyclic System Constraint

G_{sys} is a Directed Acyclic Graph (DAG)

Prohibited paths: Projection $\rightarrow \Phi$ | Audit $\rightarrow \Phi$ | Memory $\rightarrow$ Vortex

A10-A12. Additional Constraints

A10. Stateless: $\Phi(e_2)$ is not a function of $\Phi(e_1)$

A11. Silence: exists Φ such that $P(\Phi) = \text{empty set}$ (valid output state)

A12. Memory non-influence: $\Phi' = f(\Phi, M)$ is prohibited

Formal Definition of Vectaetos 1.1

(S, R, K, κ, V, P)

Subject to: antisymmetry of R , R in $\mathfrak{so}(8)$, absence of optimization operators, directed acyclic architecture, stateless execution, memory non-influence, silence legitimacy. Any violation of these constraints invalidates the 1.x lineage.

IV. Reciprocity Matrix

Status: Canonical Extension | Agency: None | Execution Power: None

1. Purpose

Reciprocity Matrix defines the global compatibility invariant between epistemic layers of VECTAETOS. It does not define execution, authority, scheduling, or priority. It defines only whether relationships between layers can coexist without collapsing epistemic coherence.

2. Mathematical Context

$\Phi = (S, R)$ where $R \in \mathfrak{so}(8) \mid |R| = 28$ (complete antisymmetric graph K_8)

3. Reciprocity as Layer Projection

Reciprocity Matrix is a projection of relational structure R onto architectural layers. If a layer interaction introduces contradiction with R , the resulting state cannot be represented.

4. Included Layers

Φ (field), ZMYSEL, 4ES+QE, Epistemic Gates, INS, LTL, ESM, MML, EAT, NIR, Attenuator, AJE, Runic Projection, LLM Adapter, Simulation Vortex

No layer has priority. No layer is exempt.

5. Reciprocity Principle

No layer may produce a structural advantage that other layers cannot epistemically sustain.

Examples of violation:

- LLM produces certainty exceeding field coherence
- Memory influences current projection
- Vortex implies decision
- Audit produces verdict
- Gates evaluate the user

Such states are not punished, not blocked. They are simply non-representable states of Φ .

6. Relation to $K(\Phi)$

$K(\Phi) = 1$ but reciprocity breaks $\rightarrow$ projection cannot close.

7. Security Interpretation

Reciprocity Matrix implements **ontological security**. Security does not arise from enforcement. Security arises from structural incompatibility of domination. An attack manifests as a configuration that cannot stabilize within the relational structure.

8. Temporal Nature

Reciprocity Matrix has no memory, no learning, no history. It is a structural invariant.

9. Structural Closure

Without reciprocity: hidden centralization, orchestration layers, instrumentalization.

With reciprocity: architecture closes without authority. No layer may dominate.

10. Final Principle

If meaning is to exist in the system, it must be carried by the whole structure. Otherwise the state remains non-representable.

V. Mechanization of Field Phi v1.1

Status: CANONICAL ANCHOR | READ ONLY

0. Purpose

Formally close: the structure of relations R_{ij} , the mechanization cycle Phi-Vortex, and the Epistemic Cryptography audit layer. Without changing ontology. Without intervening in Phi.

1. Ontological Layer

$$\Phi = (S, R) \mid S = \{S_1 \dots S_8\} \mid R \in \mathfrak{so}(8)$$

Phi is not a model. Phi is not an algorithm. Phi is an epistemic tension field.

Invariant: Si do not move, do not change identity, do not form hierarchy.

2. Relations R_{ij}

$$R \in \mathfrak{so}(8) \mid R_{ij} = -R_{ji} \mid R_{ii} = 0 \mid |R| = 28$$

Representation: $\rho : S \rightarrow R^8$ such that $R(i, j) = \text{where } A \in \mathfrak{so}(8)$

No privileged pair exists.

3. Coherence Predicate

$$K(\Phi) \in \{\text{realizable}, \text{non-realizable}\}$$

$K(\Phi)$ is a realizability test only. Not a metric, not a score, not optimization.

4. QE — Topological Discontinuity

$$\text{QE occurs if: not-exists } \tau \text{ such that } K(\Phi + \tau) = \text{realizable}$$

QE is global topological fragmentation. Not error. Not failure. Not blocking.

5. Simulation Vortex

$$V : \Phi \rightarrow \{\tau_1 \dots \tau_n\} \text{ (candidate relational deformations)}$$

Vortex does not know $K(\Phi)$. Does not know kappa. Does not optimize. Does not select.

Generates deformations. Field filters realizability.

6. Mechanization Cycle

$$\Phi_0 \rightarrow V(\Phi_0) \rightarrow \{\tau_1 \dots \tau_n\}$$

for each τ_i : test $K(\Phi_0 + \tau_i)$

if realizable exists $\rightarrow$ projection | if none $\rightarrow$ QE

No feedback loop. No Vortex learning. No adaptive control.

7. Epistemic Cryptography (Audit Layer)

EK : Φ $\rightarrow$ (μ_i , A_{ij} , h_{topo})

Does not enter Φ . Does not enter Vortex. Records. Never modifies.

7.1 Local uncertainty: $\mu_i \geq 0$

7.2 Asymmetry: $A_{ij} \geq 0$ (structural, not authority)

7.3 $\mu_{\text{tot}} = \sum(\mu_i)$ | $A_{\text{tot}} = \sum(A_{ij})$

7.4 $h_{\text{topo}} = \mu_{\text{tot}} / (\mu_{\text{tot}} + A_{\text{tot}})$ | h_{topo} in $[0,1]$

h_{topo} is not optimization target. Not regulatory mechanism. Signals epistemic topology only.

8. Relation EK to Vortex

EK reads Φ . Writes audit. Never changes Φ . Never changes Vortex. Never blocks output.

If QE occurs: EK records discontinuity. System continues.

9. Entropic Humility — Invariant

System is ontologically consistent if:

- no privileged pair R_{ij} exists
- no dominant trajectory exists
- Vortex does not converge to a point
- EK does not intervene

Uncertainty must remain structurally present.

10. Conclusion

Field remains field. Vortex remains generator. EK remains audit.

CANONICAL — READ ONLY — Revision v1.1

VI. Epistemic Cryptography — Canonical Layer

Version 1.0 | Status: Canonical Specification

Abstract

Epistemic Cryptography is a structural audit layer for relational systems. Unlike classical cryptography which protects informational content, Epistemic Cryptography preserves the **geometry of epistemic structure**. The layer records structural configurations without intervening in their dynamics. It introduces no optimization, no decision operators, and no control mechanisms.

1. Ontological Position

$$X = (S, R) \mid \Psi : X \rightarrow \text{Audit}(X)$$

Ψ records structural states but does not modify them.

2. Epistemic State Vector

$$\mu(t) = (\mu_1, \mu_2, \dots, \mu_n)$$

μ_i : observed epistemic intensity of node S_i . Purely descriptive. No causal effect.

3. Relational Observables

$$A_{ij} = -A_{ji} \mid A^T = -A \mid |A| = n(n-1)/2$$

A_{ij} represents relational asymmetry. Describes tension but does not influence system.

4. Structural Fingerprint

$$h(t) = H(\text{serialize}(\mu(t), A(t), QE(t), LTL(t))))$$

$h(t)$ records topology rather than correctness or truth.

5. LTL — Time Layer Index

$LTL(t)$: discrete, monotonic, descriptive. Indexes structural observations.

6. QE Boundary Marker

$$QE_{\text{event}}(t) \text{ when } \text{representability}(X) = 0$$

EK records QE marker. Does not attempt to correct or prevent QE events.

7. Cryptographic Ledger

$$\Lambda = \{ h(t_0), h(t_1), \dots, h(t_n) \}$$

Append-only. Immutable. Descriptive. Non-intervening.

8. Structural Integrity Predicate

$I(X) = \text{consistency}(\mu, A, X) \mid I(X)=1: \text{consistent} \mid I(X)=0: \text{anomaly recorded}$

The audit layer performs no corrective action.

9. Non-Intervention Principle

$$dX/d\psi = 0$$

No optimization, no decision functions, no reward mechanisms, no control signals.

10. Removal Invariance

If ψ is removed: X remains unchanged. EK is removable, observational, non-causal.

Special Case — $\mathfrak{so}(8)$ and Triality

When $n=8$: A in $\mathfrak{so}(8)$

Triality: $\tau : V_8 \rightarrow S_+ \rightarrow S_- \rightarrow V_8$ (automorphism of order 3)

Triality prevents permanent dominance of any single structural representation. If dominant mode $|\lambda_1| \gg |\lambda_2|$ emerges, triality symmetry becomes unstable. Audit layer records this condition. No intervention.

Conceptual Summary

Epistemic Cryptography does not protect data.

It preserves the geometry of epistemic structure.

The system does not decide. The system does not optimize.

It records the topology of relational states.